

Data pre-processing Report: Loan Approval Prediction

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1. Introduction and Business Problem

This report documents the data pre-processing pipeline applied to the Loan Prediction Dataset. The core business problem is to automate and optimize the loan approval process for financial institutions by predicting whether a loan application should be approved (Y) or rejected (N). To build an accurate machine-learning model, the raw historical data must be transformed into a clean, structured, and machine-learning-ready format. This report details every pre-processing decision, technique, and justification applied to the dataset.

2. Data Inspection and Quality Assessment

Upon loading the training dataset (614 rows, 13 columns), an initial inspection revealed several data quality issues that required immediate attention:

Irrelevant Features: The 'Loan_ID' column was identified as a unique identifier with no predictive power.

Missing Values: Significant missing data was found in 'Gender', 'Married', 'Dependents', 'Self_Employed', 'LoanAmount', 'Loan_Amount_Term', and critically, 'Credit_History'.

Data Types: The 'Dependents' column contained a mix of integers and text (e.g., '3+'), requiring type correction.

Duplicates: No exact duplicate rows were found, so no records were dropped for duplication.

3. Handling Missing Values

To retain as much data as possible while maintaining statistical integrity, the following imputation strategies were applied:

Categorical Variables ('Gender', 'Married', 'Self_Employed', 'Dependents'): Filled using the Mode (most frequent category). This is the standard approach for categorical data as it preserves the dominant distribution.

Numerical Variables (`LoanAmount`, `Loan\_Amount\_Term`): Filled using the Median . The median was chosen over the mean because financial data like loan amounts are typically right-skewed and contain extreme outliers. The median is robust against these outliers.

`Credit\_History`: Filled with the mode (1.0), as the majority of applicants in the dataset had a good credit history.

4. Feature Engineering

Feature engineering was performed to create new variables that provide stronger predictive signals to the machine learning model:

`TotalIncome`: Created by summing `ApplicantIncome` and `CoapplicantIncome`. This provides a more accurate picture of the household's total repayment capacity.

`Loan\_to\_Income\_Ratio`: Calculated as $\text{LoanAmount} / \text{TotalIncome}$. This measures the financial burden of the loan relative to the applicant's earnings, a critical metric in credit risk assessment.

`EMI` (Equated Monthly Installment): Approximated as $\text{LoanAmount} / \text{Loan_Amount_Term}$. This estimates the monthly financial strain on the applicant.

Data Type Correction: The `Dependents` column was cleaned by replacing '3+' with 3 and converting the entire column to an integer type.

5. Outlier Detection and Treatment

Financial datasets are notorious for extreme outliers (e.g., exceptionally high incomes). Boxplot visualizations confirmed severe right-skewness and outliers in `ApplicantIncome`, `CoapplicantIncome`, `LoanAmount`, and the newly engineered `TotalIncome`.

Technique Used: The Interquartile Range (IQR) method was applied.

Treatment: Instead of dropping these rows (which would result in a loss of valuable information), the outliers were capped (clipped) at the lower and upper bounds ($Q1 - 1.5 \text{ IQR}$ and $Q3 + 1.5 \text{ IQR}$). This limits the influence of extreme values on the model without deleting the records.

6. Feature Encoding

Machine learning algorithms require numerical input. The following encoding techniques were applied to categorical variables:

Binary Mapping (Label Encoding): Applied to binary categorical variables (`Gender`, `Married`, `Education`, `Self\_Employed`, and the target `Loan\_Status`). For example, Male=1, Female=0; Yes=1, No=0.

One-Hot Encoding: Applied to `Property\_Area` (Urban, Semiurban, Rural). Because this is a nominal variable with no inherent mathematical order, One-Hot Encoding prevents the model from falsely assuming that Urban > Semiurban > Rural. This created three separate binary columns.

7. Feature Scaling

Numerical features in the dataset exist on vastly different scales (e.g., Income in thousands vs. Loan Term in months). If left unscaled, algorithms that rely on distance metrics or gradient descent will be biased toward features with larger magnitudes.

Technique Used: `StandardScaler` (Standardization).

Justification: `StandardScaler` transforms the data to have a mean of 0 and a standard deviation of 1. It is highly effective for algorithms like Logistic Regression, Support Vector Machines (SVM), and Neural Networks, ensuring all features contribute equally to the model's learning process.

8. Feature Selection

To prevent multicollinearity (where features are highly correlated and confuse the model) and reduce dimensionality, a correlation heatmap was generated.

Redundant Features Removed: Because we engineered `TotalIncome`, the original `ApplicantIncome` and `CoapplicantIncome` columns became redundant and were dropped. Similarly, the original `LoanAmount` was dropped in favor of the more informative `Loan_to_Income_Ratio` and `EMI`.

This resulted in a streamlined, highly predictive feature set.

9. Conclusion

The pre-processing pipeline successfully transformed a raw, messy dataset containing missing values, outliers, and mixed data types into a pristine, machine-learning-ready dataset. The final dataset contains scaled, encoded, and engineered features with zero missing values, making it perfectly optimized for training robust classification models in the next phase of the project.